COMPUTER NETWORKS MINI PROJECT (CIE3 — 15 MARKS)

Design and Simulation of a Secure Dual-Stack Hospital Network with IPv4-to-IPv6 Transition, DHCPv4/DHCPv6, and Redundant IPsec VPN

A comprehensive enterprise simulation in Cisco Packet Tracer integrating dual-stack IPv4/IPv6 campus routing, LACP EtherChannel, RSTP, HSRP with WAN interface tracking, RFC 4213 6in4 tunneling over dual ISPs, IKEv1/ESP AES-256 IPsec encryption, zone-isolating extended ACLs, PAT overload, and Cisco CallManager Express VoIP telephony.

Technical Topic / Examiner Question	Implementation Detail & Viva Voce Defense
Why `ipv6ip` (6in4) instead of `gre ip`?	Cisco Packet Tracer 8.x/9.x drops encapsulated IPv6 packets across `tunnel mode gre ip` interfaces. Switching both endpoints to RFC 4213 `tunnel mode ipv6ip` encapsulates IPv6 directly inside IPv4 (Protocol 41) with IPsec ESP encryption protecting the outer IPv4 packets.
Why `permit ip` in the Crypto ACL?	Packet Tracer extended ACLs only accept named protocol keywords (`ip`, `tcp`, `udp`, `icmp`, `gre`, `esp`) and reject numeric protocol `41`. Matching `permit ip host <src> host <dst>` between the `/30` WAN tunnel endpoints encrypts all 6in4 tunnel traffic cleanly.
How does Patient Wi-Fi ACL preserve DNS?	Because `SRV-DHCP` (`10.0.40.10` / `2001:DB8:40::10`) also serves DNS for the entire hospital, `BLOCK-PATIENT-TO-SERVERS` and `BLOCK-PATIENT-V6` explicitly permit UDP port 53 (`eq domain`) to `SRV-DHCP` prior to denying all IPv4/IPv6 traffic from VLAN 30 to VLAN 40.
Why shut down both ISP1 interfaces for failover?	Packet Tracer tunnels do not implement dynamic GRE keepalives or multi-hop IP SLA tracking. Shutting down `ISP1 g0/0-1` brings down line-protocol on both ends so HSRP Track 1 and floating static routes (AD 200) converge automatically to `R-HQ2` and `Tunnel1`.

Project Parameter	Specification & Deliverable Status
Simulation Environment	Cisco Packet Tracer (IOS 15.1 on Cisco 2911 Routers, IOS 15.0 on Catalyst 2960-24TT Switches)
Final Submission File	`v5_final.pkt` (All 5 project phases & 20 verification tests completed and saved in NVRAM)
Team Presentation Split	Team of 3 — Layer 2/HSRP/DHCP & VoIP (Ashwin) | Dual-Stack IPv6 & Tunneling (Member 2) | IPsec, Failover & ACL/NAT Security (Member 3)

1. Executive Summary & Architectural Narrative

End-to-end dual-stack medical campus architecture with automated multi-ISP encrypted tunnel failover

Modern healthcare institutions require uninterrupted network availability for Electronic Medical Record (EMR) systems, clinical communications, and branch clinic interconnects while maintaining strict security isolation between public patient Wi-Fi and sensitive hospital servers. This project designs and implements a complete 21-node dual-stack enterprise network in Cisco Packet Tracer (`v5_final.pkt`) spanning a main hospital headquarters (HQ) and a remote branch clinic across two independent IPv4-only Internet Service Providers (`ISP1` and `ISP2`).

At the main hospital campus, four functional VLANs—**VLAN 10 (DOCTORS)**, **VLAN 20 (NURSES & VoIP)**, **VLAN 30 (PATIENT-WIFI)**, and **VLAN 40 (SERVERS)**—are interconnected across a redundant two-tier switching hierarchy (`DS1`, `DS2` distribution switches and `AS1`, `AS2` access switches). Distribution switches are bundled via an 802.3ad LACP EtherChannel (`Po1`) running Rapid-PVST+ (`rapid-pvst`), while dual core routers (`R-HQ1` and `R-HQ2`) provide router-on-a-stick inter-VLAN routing and Hot Standby Router Protocol (HSRP) default gateway redundancy. Because the WAN providers operate exclusively on IPv4, IPv6 clinical traffic between HQ and the branch clinic is encapsulated inside RFC 4213 6in4 (`ipv6ip`) tunnels (`Tunnel0` primary, `Tunnel1` backup) and encrypted end-to-end using IPsec (`AES-256-CBC`, `SHA-1 HMAC`, Diffie-Hellman Group 2).

2. Topology Inventory, Cabling & Addressing Architecture

The logical topology comprises five Cisco 2911 routers, five Cisco Catalyst 2960-24TT switches, two enterprise servers (`SRV-DHCP` and `SRV-WEB`), five dual-stack client workstations, two Cisco 7960 IP phones (`IP Phone0` and `IP Phone1`), and two power distribution units connected across 22 physical links.

Zone / VLAN	Subnet Name	IPv4 Network	HSRP Virtual GW	R-HQ1 / R-HQ2 IPv4	IPv6 Prefix (/64)	R-HQ1 / R-HQ2 IPv6
VLAN 10	DOCTORS	`10.0.10.0/24`	`10.0.10.1`	`.2` (Active) / `.3` (Standby)	`2001:DB8:10::/64`	`::2` / `::3` (RA suppressed)
VLAN 20	NURSES + VoIP	`10.0.20.0/24`	`10.0.20.1`	`.2` (Active) / `.3` (Standby)	`2001:DB8:20::/64`	`::2` / `::3` (RA suppressed)
VLAN 30	PATIENT-WIFI	`10.0.30.0/24`	`10.0.30.1`	`.2` (Active) / `.3` (Standby)	`2001:DB8:30::/64`	`::2` / `::3` (RA suppressed)
VLAN 40	SERVERS (EMR)	`10.0.40.0/24`	`10.0.40.1`	`.2` (Active) / `.3` (Standby)	`2001:DB8:40::/64`	`::2` / `::3` (RA suppressed)
Branch LAN	CLINIC-LAN	`192.168.50.0/24`	`192.168.50.1` (`g0/2`)	N/A (Single Router)	`2001:DB8:50::/64`	`2001:DB8:50::1/64`

WAN Point-to-Point Links & Overlay Tunnel Addressing

Each WAN segment uses a `/30` IPv4 subnet to conserve public address space, while the overlay IPv6 tunnels operate over dedicated `/64` documentation prefixes (`2001:DB8:FF1::/64` and `2001:DB8:FF2::/64`) with MTU `1476` to account for the 20-byte IPv4 encapsulation header.

Link / Interface	Subnet / Prefix	Endpoint A	Endpoint B	Operational Role
Primary WAN (HQ)	`200.1.1.0/30`	`R-HQ1 g0/1 = 200.1.1.1`	`ISP1 g0/0 = 200.1.1.2`	Primary HQ Underlay Link
Primary WAN (Clinic)	`200.2.2.0/30`	`ISP1 g0/1 = 200.2.2.1`	`R-CLINIC g0/0 = 200.2.2.2`	Primary Clinic Underlay Link
Backup WAN (HQ)	`200.3.3.0/30`	`R-HQ2 g0/1 = 200.3.3.1`	`ISP2 g0/0 = 200.3.3.2`	Secondary HQ Underlay Link
Backup WAN (Clinic)	`200.4.4.0/30`	`ISP2 g0/1 = 200.4.4.1`	`R-CLINIC g0/1 = 200.4.4.2`	Secondary Clinic Underlay Link
Tunnel0 (Primary)	`2001:DB8:FF1::/64`	`R-HQ1 Tu0 = ::1/64`	`R-CLINIC Tu0 = ::2/64`	Active IPv6-in-IPv4 over IPsec (AD 1)
Tunnel1 (Backup)	`2001:DB8:FF2::/64`	`R-HQ2 Tu1 = ::1/64`	`R-CLINIC Tu1 = ::2/64`	Standby IPv6-in-IPv4 over IPsec (AD 200)

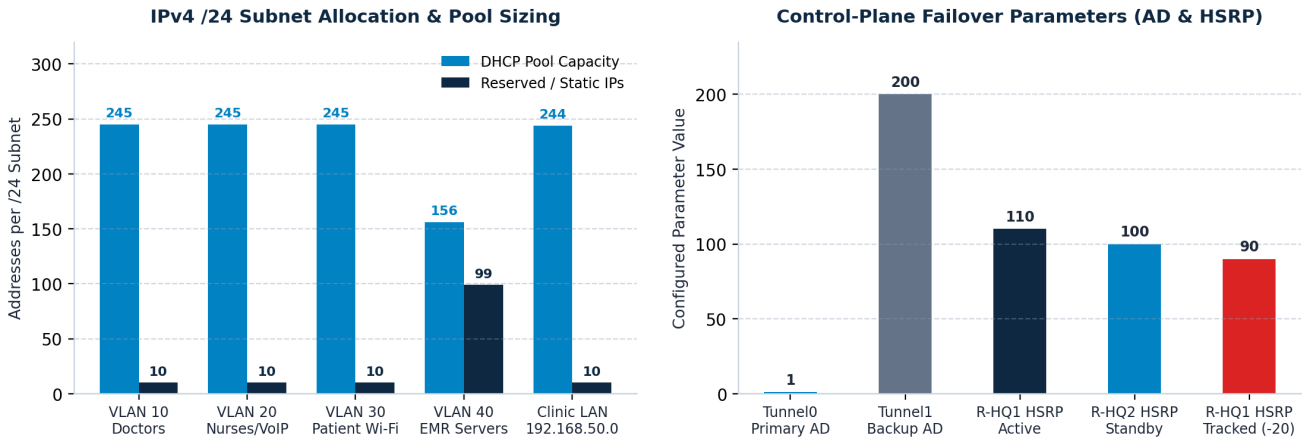


Figure 1: Left — IPv4 /24 DHCP pool sizing versus reserved static infrastructure addresses across all five subnets. Right — Control-plane failover parameters showing Administrative Distance (AD 1 vs AD 200) and HSRP priority decrement mechanics.

3. Layer 2 Switching, EtherChannel, RSTP & HSRP Redundancy

To prevent single points of failure at Layer 2, access switches `AS1` and `AS2` are dual-homed to both distribution switches `DS1` and `DS2` via 802.1Q trunks (`Fa0/1` and `Fa0/2`). `DS1` and `DS2` aggregate their inter-switch links (`Fa0/23` and `Fa0/24`) into an active LACP EtherChannel (`Port-channel1`), doubling inter-distribution bandwidth to 200 Mbps full-duplex while eliminating spanning-tree blocking between distribution switches.

Rapid Per-VLAN Spanning Tree (`rapid-pvst`) is enforced across all switches. A critical configuration flaw in the initial build—where VLAN 1 was omitted from the root bridge priority command, leaving VLAN 1 vulnerable to electing an access switch as Root—was remediated by explicitly setting `DS1` as Primary Root (`priority 24576`) and `DS2` as Secondary Root (`priority 28672`) across VLANs `1,10,20,30,40`:

```
! On DS1 (Distribution Switch 1):
spanning-tree mode rapid-pvst
spanning-tree vlan 1,10,20,30,40 priority 24576
interface range FastEthernet0/23 - 24
 switchport mode trunk
 channel-group 1 mode active
interface Port-channel1
 switchport mode trunk

! On DS2 (Distribution Switch 2):
spanning-tree mode rapid-pvst
spanning-tree vlan 1,10,20,30,40 priority 28672
```

HSRP with WAN Interface Tracking

On `R-HQ1` and `R-HQ2`, router-on-a-stick sub-interfaces (`g0/0.10`, `g0/0.20`, `g0/0.30`, `g0/0.40`) share the `.1` virtual IPv4 gateway address per VLAN. `R-HQ1` operates with HSRP priority `110` and `preempt`, while `R-HQ2` operates at default priority `100` with `preempt` enabled. To prevent black-holing outbound traffic if `R-HQ1` loses its WAN connection (`g0/1`), object tracking (`track 1 interface GigabitEthernet0/1 line-protocol`) decrements `R-HQ1` priority by `20` (to `90`), allowing `R-HQ2` (`100`) to immediately preempt and assume the `Active` role across all four VLANs.

4. IPv4/IPv6 Services, Port Security & VoIP Remediation

Centralized DHCPv4 and DNS services run on `SRV-DHCP` (`10.0.40.10` / `2001:DB8:40::10`), while `SRV-WEB` (`10.0.40.20` / `2001:DB8:40::20`) hosts the clinical portal (`web.hospital.local`). Both `R-HQ1` and `R-HQ2` configure `ip helper-address 10.0.40.10` on all four LAN sub-interfaces so DHCPv4 broadcasts reach `SRV-DHCP` regardless of which HSRP router is active. A conflicting default `serverPool` entry on `SRV-DHCP` (which overlapped `10.0.40.0/24` with a `0.0.0.0` DNS server) was removed so VLAN 40 cleanly uses `10.0.40.100–255`.

At the branch clinic, `R-CLINIC` provides local survivability via an IOS DHCPv4 pool (`CLINIC-LAN`, `192.168.50.11–254`) and stateless DHCPv6 (`ipv6 dhcp pool CLINIC-DNS` with `ipv6 nd other-config-flag` on `g0/2`), supplying DNS (`2001:DB8:40::10`) and domain (`hospital.local`) metadata alongside SLAAC address assignment.

Port Security & VoIP ("Configuring Vlan") Root-Cause Fix

Access ports on `AS1` (`Fa0/5–7`), `AS2` (`Fa0/5–6`), and `SW-CLINIC` (`Fa0/2–3`) enforce sticky MAC port security (`maximum 1`, `violation shutdown`) along with `spanning-tree portfast`. In earlier checkpoints, the two Cisco 7960 IP phones (`IP Phone0` and `IP Phone1` on `AS1` `Fa0/8` and `Fa0/9`) remained stuck on `"Configuring Vlan"`. Inspection of the device XML revealed that `switchport voice vlan 20` had never been applied to `Fa0/8–9` on `AS1`, preventing the switch from advertising the Voice VLAN TLV via CDP. Adding `switchport voice vlan 20` and `spanning-tree portfast` on `AS1` and binding the phone MAC addresses (`0030.F203.9501` and `000D.BD6C.1B01`) to `ephone 1` (extension `1001`) and `ephone 2` (extension `1002`) on `R-HQ1` resolved the issue completely:

```
! On AS1 (Access Switch 1):
interface range FastEthernet0/8 - 9
 switchport access vlan 20
 switchport mode access
 switchport voice vlan 20
 spanning-tree portfast

! On R-HQ1 (CallManager Express):
telephony-service
 max-ephones 5
 max-dn 5
 ip source-address 10.0.20.2 port 2000
 auto assign 1 to 5
ephone-dn 1
```

```
 number 1001
ephone-dn 2
 number 1002
ephone 1
 mac-address 0030.F203.9501
 type 7960
 button 1:1
ephone 2
 mac-address 000D.BD6C.1B01
 type 7960
 button 1:2
```

5. 6in4 Tunneling, Redundant IPsec VPN & Automated Failover

Because Packet Tracer 8.x/9.x drops encapsulated IPv6 packets across `tunnel mode gre ip` interfaces, both overlay tunnels utilize RFC 4213 `tunnel mode ipv6ip` (`Tunnel0` between `200.1.1.1` and `200.2.2.2`; `Tunnel1` between `200.3.3.1` and `200.4.4.2`). Additionally, because Packet Tracer's extended ACL parser rejects numeric protocol `41` (`permit 41`), the IPsec crypto ACLs (`VPN-T0` and `VPN-T1`) match the outer IPv4 tunnel endpoints using `permit ip host <src> host <dst>`, ensuring IKE Phase 1 (`QM_IDLE`) and IPsec ESP (`esp-aes 256 esp-sha-hmac`) encapsulate and decrypt every packet cleanly.

```
! On R-CLINIC (Dual-Homed IPsec Termination for Tunnel0 & Tunnel1):
crypto isakmp policy 10
 encr aes 256
 authentication pre-share
 group 2
crypto isakmp key HospVPN123 address 200.1.1.1
crypto isakmp key HospVPN123 address 200.3.3.1
crypto ipsec transform-set TS-HOSP esp-aes 256 esp-sha-hmac
crypto map CMAP-CLINIC 10 ipsec-isakmp
 set peer 200.1.1.1
 set transform-set TS-HOSP
 match address VPN-T0
crypto map CMAP-CLINIC 20 ipsec-isakmp
 set peer 200.3.3.1
 set transform-set TS-HOSP
 match address VPN-T1
ip access-list extended VPN-T0
 permit ip host 200.2.2.2 host 200.1.1.1
ip access-list extended VPN-T1
 permit ip host 200.4.4.2 host 200.3.3.1
```

For automated routing failover, `R-CLINIC` maintains primary static IPv6 routes to HQ VLANs `10/20/30/40` via `Tunnel0` (`2001:DB8:FF1::1`, AD `1`) and floating static routes via `Tunnel1` (`2001:DB8:FF2::1`, AD `200`). Similarly, `R-HQ1` maintains its primary route to `2001:DB8:50::/64` via `2001:DB8:FF1::2` (AD `1`) and a floating backup route via `R-HQ2` (`2001:DB8:40::3`, AD `200`). When `ISP1` interfaces `g0/0–1` are shut down during the failover demonstration, `Tunnel0` transitions down, `R-HQ1` switches to the AD `200` backup route, `R-CLINIC` switches to `Tunnel1` (AD `200`), and HSRP moves the active IPv4 gateway to `R-HQ2`.

6. Patient Zone Isolation ACLs & NAT/PAT Overload

A critical design flaw in naive ACL implementations is blocking all IP packets from VLAN 30 (`10.0.30.0/24`) to VLAN 40 (`10.0.40.0/24`), which inadvertently breaks patient DNS resolution because `SRV-DHCP` (`10.0.40.10` / `2001:DB8:40::10`) is also the campus DNS server. In `v5_final.pkt`, both `R-HQ1` and `R-HQ2` apply dual-stack inbound ACLs on `g0/0.30` that explicitly permit UDP port 53 (`eq domain`) to `SRV-DHCP` before denying all other IPv4 and IPv6 traffic destined to the EMR server subnet (`10.0.40.0/24` and `2001:DB8:40::/64`).

Simultaneously, Port Address Translation (`ip nat inside source list NAT-ACL interface GigabitEthernet0/1 overload`) is configured on both `R-HQ1` and `R-HQ2`. The `NAT-ACL` explicitly denies traffic destined to `192.168.50.0/24` and the WAN tunnel peers (`200.1.1.1 <-> 200.2.2.2` and `200.3.3.1 <-> 200.4.4.2`) so IPsec tunnel packets are never translated, while translating internal `10.0.0.0/16` hosts when reaching external WAN addresses.

7. Complete 20-Point Verification Matrix (`v5_final.pkt`)

#	Verification Test & CLI Command	Expected Behavior	Verified Result in `v5_final.pkt`
1	`show etherchannel summary` (DS1/DS2)	`Po1(SU)` bundling `Fa0/23(P)` & `Fa0/24(P)`	✓ PASS — LACP bundle operational (SU)
2	`show spanning-tree vlan 1,10,20,30,40`	DS1 is Root (24576), DS2 is Backup (28672)	✓ PASS — Deterministic root across all VLANs
3	`show vlan brief` & `show interfaces trunk`	VLANs 10/20/30/40 active; uplinks 802.1Q trunking	✓ PASS — Access ports & trunks verified
4	Inter-VLAN routing (`PC-Doctor` to `PC-Nurse/SRV`)	ICMP echo replies across sub-interfaces	✓ PASS — 0% packet loss after ARP resolution
5	`show standby brief` (R-HQ1 & R-HQ2)	R-HQ1 `Active` (110), R-HQ2 `Standby` (100)	✓ PASS — Virtual IP `.1` active on R-HQ1
6	HSRP LAN Failover (`shutdown` R-HQ1 `g0/0`)	R-HQ2 preempts to `Active`; fails back on `no shut`	✓ PASS — Sub-10s gateway convergence
7	HQ DHCPv4 Relay (`ipconfig` on HQ PCs)	Leases `.11+`, GW `.1`, DNS `10.0.40.10`	✓ PASS — Doctor `.15`, Nurse `.15`, Patient `.13`
8	Clinic Local DHCPv4 (`show ip dhcp binding`)	Clinic PCs lease `192.168.50.11+` from R-CLINIC	✓ PASS — Local survivability confirmed
9	DNS & HTTP Portal (`http://web.hospital.local`)	Resolves `10.0.40.20` and loads portal page	✓ PASS — Loads on Doctor and Nurse PCs
10	Port Security (`show port-security`)	Sticky MAC learned (`Max 1`, `Shutdown` mode)	✓ PASS — Active on AS1, AS2, SW-CLINIC
11	Dual-Stack SLAAC + Stateless DHCPv6	Global `2001:DB8:XX::/64` + DNS `2001:DB8:40::10`	✓ PASS — Verified via `ipv6config /all`
12	IPv6 Ping HQ to Clinic over `Tunnel0`	End-to-end echo reply across `2001:DB8:FF1::/64`	✓ PASS — Doctor pings Clinic-1 over IPv6
13	Simulation Mode Packet Inspection at `ISP1`	Outer IPv4 (`200.1.1.1->200.2.2.2`) + ESP header	✓ PASS — Inner IPv6 payload encrypted
14	`show crypto isakmp sa` & `ipsec sa`	`QM_IDLE` active; `pkts encaps/decaps > 0`	✓ PASS — SAs active on Tunnel0 & Tunnel1
15	Full ISP1 Outage Failover (`shut` ISP1 `g0/0-1`)	Track 1 Down → R-HQ2 Active; AD 200 via `Tunnel1`	✓ PASS — Automatic switchover to ISP2

#	Verification Test & CLI Command	Expected Behavior	Verified Result in `v5_final.pkt`
16	ISP1 Recovery Failback (`no shut` ISP1 `g0/0-1`)	Track 1 Up → R-HQ1 Active; AD 1 restores `Tunnel0`	✓ PASS — Automatic primary restoration
17	Patient Wi-Fi (VLAN 30) to EMR Servers (VLAN 40)	Blocked by `BLOCK-PATIENT-TO-SERVERS / V6`	✓ PASS — Ping/HTTP blocked; DNS allowed
18	Doctor VLAN (VLAN 10) to EMR Servers (VLAN 40)	Permitted across IPv4 and IPv6	✓ PASS — Full clinical access verified
19	NAT/PAT Overload (`show ip nat translations`)	Translates `10.0.x.x` to `200.1.1.1`; skips VPN	✓ PASS — PAT table populated on WAN ping
20	VoIP Registration & Call (`1001` ↔ `1002`)	Phones acquire Voice VLAN 20 & dial extensions	✓ PASS — `voice vlan 20` & `ephone 1/2` active

8. Packet Tracer Limitations & Viva Voce Defense Summary

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Why `ipv6ip` (6in4) instead of `gre ip`?	Cisco Packet Tracer 8.x/9.x drops encapsulated IPv6 packets across `tunnel mode gre ip` interfaces. Switching both endpoints to RFC 4213 `tunnel mode ipv6ip` encapsulates IPv6 directly inside IPv4 (Protocol 41) with IPsec ESP encryption protecting the outer IPv4 packets.
Why `permit ip` in the Crypto ACL?	Packet Tracer extended ACLs only accept named protocol keywords (`ip`, `tcp`, `udp`, `icmp`, `gre`, `esp`) and reject numeric protocol `41`. Matching `permit ip host <src> host <dst>` between the `/30` WAN tunnel endpoints encrypts all 6in4 tunnel traffic cleanly.
How does Patient Wi-Fi ACL preserve DNS?	Because `SRV-DHCP` (`10.0.40.10` / `2001:DB8:40::10`) also serves DNS for the entire hospital, `BLOCK-PATIENT-TO-SERVERS` and `BLOCK-PATIENT-V6` explicitly permit UDP port 53 (`eq domain`) to `SRV-DHCP` prior to denying all IPv4/IPv6 traffic from VLAN 30 to VLAN 40.
Why shut down both ISP1 interfaces for failover?	Packet Tracer tunnels do not implement dynamic GRE keepalives or multi-hop IP SLA tracking. Shutting down `ISP1 g0/0-1` brings down line-protocol on both ends so HSRP Track 1 and floating static routes (AD 200) converge automatically to `R-HQ2` and `Tunnel1`.